

Judging

Criterion	What it covers
Inspiration	A clear problem with real impact
Technology	Feasibility, quality, how you use AI
Design	Usability and originality
Presentation	Clear, structured, within 5 minutes

Local winners go through to national judging.

What scores well

Judges reward execution over novelty. Working software aimed at a specific group beats a clever idea that only half runs. A plain LLM wrapper no longer stands out. Health and education chatbots are the most common entries, so they are the hardest lane to win in.

One thing separates entries reliably: producing a number that maps to an official UN indicator. It gives judges something concrete to check.

Context

Only 16% of SDG targets are projected to be met by 2030. The UN's own count puts 36% of 139 targets on track or progressing moderately. Four years left.

Two of the three problems suggested in the event PDF — Facilities Helpdesk and Peer Support — are chatbot-shaped. Choosing one means competing mostly on presentation.

Idea 1 — Cold-chain monitor for last-mile clinics

Problem Vaccines and insulin spoil quietly when a fridge fails or a transport box warms above 8 °C. Rural clinics often find out only when the stock is already unusable.

SDG SDG 3 (Good Health). Protects the potency of health supplies reaching underserved patients. Indicator 3.b.1, vaccine coverage.

Worth knowing first

This device already exists commercially. Nexleaf ColdTrace X is WHO-prequalified and does everything an ESP32 rig would: ± 0.5 °C accuracy, BLE sensors, cellular or WiFi sync, five-year warranty, and a backend running 15,000+ devices. Berlinger Fridge-tag is the standalone equivalent.

But research on why cold chains fail points somewhere else. In one field study, 91.7% of facilities had no cold chain policy and none had an equipment contingency plan. The alert fires and nobody knows who is meant to act. That gap is open.

Approach

Keep the ESP32 and temperature sensor, but treat it as the input, not the product. The product is what happens after an excursion is detected.

1. Dose disposition. The system reads the excursion — how long, how hot or cold, which fridge — and checks it against vaccine stability rules to recommend use, quarantine, or discard for each product, showing its reasoning. Today this decision is made manually.
2. Escalation. Identifies who is on call, whether the immunisation coordinator has been contacted, and drafts the incident record.
3. Tamper-evident log. Events are hash-stamped so a flagged batch cannot be quietly un-flagged.

New Zealand angle

The local process is CCA (Cold Chain Accreditation), run by the Ministry of Health and valid up to three years. Pharmacies are audited separately by Medicines Control.

The 2017 National Standards are under review by Health NZ right now.

Current interim rule: fridges over ten years old are allowed only with continuous real-time monitoring and out-of-hours alarming. That is existing demand for this, today.

Freezing, not overheating, causes most vaccine damage in NZ.

New Zealand's cold chain programme cut vaccine wastage from 17% to 2% between 2002 and 2008, saving an estimated \$4M a year.

Build and demo

Offline-first Buffers locally, syncs on reconnect.

Why 48 hours works The firmware is a read-sensor, append-log loop. The web layer is standard CRUD with a time-series chart. The decision layer is a prompt over a rules table, not a model you have to train.

Demo Put the probe in a cup of ice and pull it out mid-pitch. The dashboard flags the excursion, then the system produces a disposition recommendation and a drafted incident report on screen.

Risks

Venue WiFi is unreliable. Use BLE-to-phone or a local access point as the main sync path.

Frame output as decision support pending coordinator sign-off, not as a clinical ruling.

Record a video of the excursion firing as a fallback.

Solid if you pitch the decision layer. Weak if you pitch the logger.

Idea 2 — Surplus matching for growers

Problem Produce is culled before packing for cosmetic reasons and sold by the bin, not the bag. It never enters a retail inventory system, so no existing food-waste platform can see it. Meanwhile household food waste in NZ is already falling — 12.2% to 10.9% of food bought, \$3.2B to \$3.0B — so the consumer end of the problem is shrinking while this end is not.

SDG SDG 12.3 (halve food waste) and SDG 2.1 (end hunger). Two indicators, one build.

The numbers

NZ Food Network and Lineage redirected over 1,000 pallets — about 635,000 kg — of frozen vegetables rejected only for cosmetic irregularities.

One in three NZ households faced food insecurity in the past year (Hunger Monitor 2025, published March 2026).

The burden is uneven: Pacific peoples 64%, Māori 51%, ages 18–24 50%.

Only 44% of food-insecure households have ever accessed food relief.

Why this is open

NZ food rescue matching is route-based and manual. NZFN's stated approach for bulk surplus is to act as a single point of contact — a human broker. Regular donors sit on fixed recurring routes. The existing apps solve adjacent problems: Olio is neighbour-to-neighbour, Too Good To Go and Flashfood work off retail shelves, Spoiler Alert handles packaged goods and inventory. Imperfect Foods and Misfits Market turned cosmetic rejects into a subscription box rather than a matching layer. None of them can see stock that is culled before it becomes inventory.

Approach

4. A grower photographs a bin. A vision model grades the reject class and estimates volume. Manual data entry is what stops growers participating, so removing it matters more than it sounds.
5. A matcher scores allocations against shelf life, cold storage availability, hub capacity, travel time, and need severity. This is a scoring function, not a chatbot.
6. An impact ledger tracks kg diverted, meal equivalents, and CO₂e avoided, mapped to SDG 12.3.

Domain detail and scale

The Food Act 2014 excludes food with compromised sanitary seals unless it is heated. Naming a real constraint like this shows you looked at the sector.

64 food hubs, 2,000+ frontline groups, and roughly 28 million meal equivalents a year are already moving. You would be adding a layer to an existing network rather than building one.

Build and demo

Why 48 hours works Vision grading is one API call with a rubric. A greedy allocator demos the same as an optimal one at this scale. The dashboard is CRUD, a map, and a chart.

Demo Photograph a crate of misshapen carrots. It gets graded, the matcher allocates across three mock hubs, and the ledger updates. Then mark a hub's chiller offline and let it reallocate.

Risks

A two-sided marketplace with thin supply fails. Pitch it as a tool for one broker who already has both sides, not as a marketplace.

Position it as complementary to NZFN rather than competing.

The best-balanced option: local data, a real gap, AI that is not a chatbot, two indicators, and a mechanism that works beyond NZ.

Idea 3 — Disability funding reassessment tool

Problem NZ's disability support funding is changing now, and the people affected cannot work out what it means for them. From 1 April 2026 existing NASC users get a set budget with a limit. From 1 October 2026 reassessments run under the new system. Budgets are based on past spending, so anyone who underspent during a restrictive period risks being held to that lower figure.

SDG SDG 10.2 (social and economic inclusion regardless of disability).

Why the timing matters

Waikato is an EGL site. EGL Waikato and Mana Whaikaha whānau keep direct funding. The event is inside the affected region and the judges are local.

Reassessments start two months after the hackathon.

The policy is actively contested. Blind Low Vision NZ has flagged clauses 7 and 8 of the Disability Support Services Bill 2026 as a step away from EGL principles, framing support as “within available funding” and putting family and community ahead of State support. CCS Disability Action says Budget 2026 squeezes community-based support while protecting residential care.

Approach

7. Budget simulator. Enter current supports, model the allocation under the new past-spending rule, and show the difference — flagging where the old March 2024 purchasing restrictions may have artificially lowered someone's baseline. That is the specific harm advocates are warning about.
8. Evidence pack generator. Retrieval over the National Standards, the February 2026 fact sheet, and EGL principles, producing a reassessment submission with citations.
9. Plain-language layer. Turns clause text into readable summaries, and works the other way too — putting a whānau's account into the assessor's terms.

Accessibility

This is the one idea where accessibility is the product rather than a requirement to satisfy. WCAG 2.2 AA, screen-reader first, fully keyboard navigable. You have an accessibility skill installed with reference material.

Build and demo

Why 48 hours works The simulator is arithmetic over a rules table. The evidence pack is retrieval over roughly 40 pages of public PDFs. No training, scraping, or authentication.

Demo Two personas side by side — one whose budget holds, one whose baseline was suppressed by the 2024 rules. Same tool, different outcomes, both explained. Then generate the second persona's evidence pack.

Risks

You are modelling contested policy. Label outputs as indicative, not determinations, and cite every rule to its source.

Use synthetic personas only.

Get the rules right or do not ship. A wrong number here affects real people, and saying so in the pitch is worth more than hiding it.

Strongest local resonance, and the only idea where the accessibility brief and the technical work are the same thing.

Idea 4 — E-waste routing

Problem New Zealand produces about 99,000 tonnes of e-waste a year under voluntary schemes only. It is the OECD outlier — producer participation is not mandatory and there is no extended producer responsibility framework, unlike every other OECD country. The AS/NZS recycling standards exist but are not binding.

SDG SDG 12.5 (reduce waste generation). Indicator 12.5.1 is the national recycling rate, which NZ largely cannot report — and that is part of the story.

The gap

E-waste was designated a priority product requiring mandatory stewardship in 2020. TechCollect NZ ran the co-design between 2020 and 2023. As of mid-2026 the regulations are still not in force and no

implementation date is set. All six 2020 priority product groups remain in consultation or drafting. Six years, and 99,000 tonnes a year continues to flow.

Approach

10. Drop-off guidance. Photograph a device; a vision model identifies the type and hazardous components, then finds the nearest point that actually accepts it. Right now this is confusing enough that people give up and landfill it.

11. Repair before recycle. Route repairable devices to repair cafés and refurbishers first. Recycling is the last step in the waste hierarchy, but most tools skip straight to it.

12. Reporting ledger. Aggregate anonymised flows into the shape a regulated scheme will eventually require, so the data model exists when regulations land.

Build and demo

Low bandwidth An SMS path that assumes no smartphone makes it usable in places where informal collectors do this work, which is where 12.5 is weakest globally.

Why 48 hours works Vision identification is one API call. The list of accepting points is small enough to assemble by hand. The ledger is a schema and a chart.

Demo Photograph an old phone and a dead microwave. One routes to repair, one to a named recycler, and one is refused by the voluntary scheme entirely — that refusal gets logged as evidence of the policy gap.

Risks

Frame the regulatory angle as a scheme-ready data model, not a claim to influence policy.

Curate the accepting-points list manually rather than building a scraper.

Best reporting story of the five. Weakest emotional hook, so the pitch has to carry it.

Idea 5 — Neighbourhood SDG indicator generator

Problem The main gap in SDG 11 is measurement. Fewer than a third of sustainable-cities indicators have data at all, and coverage is worst at neighbourhood level and for vulnerable groups. HLPF 2026 called for urban data systems as a public good.

SDG SDG 11.7, access to safe and inclusive public space. Indicator 11.7.1 is the average share of built-up area that is open public space.

The numbers

Access to open public space fell from 48.0% to 45.9% between 2020 and 2025 across 414 cities in 126 countries. It is going backwards.

About 1.1 billion people live in slums or slum-like conditions, with another 2 billion projected over 30 years.

Approach

13. Compute rather than survey. Free satellite imagery, OpenStreetMap, and a segmentation model estimate open public space share for a neighbourhood, output in the form UN-Habitat's methodology expects.

14. Ground-truth layer. Residents contribute phone photos and a vision model checks whether mapped public space is actually accessible — fenced, locked, paved over, or usable. Satellites cannot see this.

15. Validate publicly. Run it on a city UN-Habitat has already measured and show your estimate next to theirs. Judges can check you on the spot, which few teams offer.

Build and demo

Why 48 hours works A pre-trained segmentation model, OpenStreetMap via the Overpass API, one city, one indicator. The discipline is keeping it to exactly that.

Demo Run it on a Hamilton suburb and show the computed figure. Then show the resident photo layer downgrading a park that turns out to be fenced off, and explain why the satellite reading was wrong.

Risks

Highest technical risk here. Segmentation quality is unpredictable and hard to debug late on day two.

Pre-compute the demo results and keep a cached fallback.

Skip this unless someone on the team has done geospatial or computer vision work before.

Highest technical ceiling, highest chance of a broken demo.

Idea 6 — Patient health literacy and informed consent

Problem Discharge notes, surgical consent forms, and diagnostic reports are written in clinical language. Patients sign and leave without understanding what was decided, what to watch for, or what happens next.

SDG 3 (Good Health), and SDG 10.2 where low literacy tracks with ethnicity and deprivation.

Indicator 3.8.1, coverage of essential health services — care a patient cannot act on is not effective coverage.

The numbers

More than 56% of New Zealand adults — over 1.6 million people — cannot obtain and understand basic health information well enough to make informed decisions.

Around 90% of Pacific people aged 15 and over have low health literacy. Māori and Pacific rates are lower than non-Māori, though the largest group by headcount is Pākehā.

16.1% of acute medical admissions among over-65s are readmitted within 30 days, with higher risk for Māori, Pacific people, and those in deprived areas.

Two corrections to the framing

Both of these make the pitch stronger, not weaker.

It is a legal right, not a service improvement. Right 5 of the Code of Health and Disability Services Consumers' Rights gives every consumer the right to effective communication “in a form, language, and manner that enables the consumer to understand the information provided,” including a competent interpreter where necessary and reasonably practicable. Right 6 covers being fully informed. A consent form the patient did not understand is arguably not compliant. That reframes the tool from helpful to required.

Avoid the word compliance. New Zealand research consistently frames post-discharge failure as system-side: inadequate support, medication problems, unpreparedness for community living, limited whānau support, and poor access to culturally responsive services. Readmission signals unmet need rather than patient fault. Saying “poor patient compliance” in front of NZ health judges will cost you.

What already exists

Epic's patient-facing assistant, Emmie, already answers patient questions inside MyChart and produces plain-language summaries of lab and procedure results. Do not pitch a generic medical-jargon translator as though nothing exists.

Two things remain genuinely open. Epic has no shipped feature that rewrites the full inpatient discharge summary, and New Zealand's public hospitals do not run on Epic, so none of that reaches a patient at Waikato Hospital. The evidence is also strong: an LLM-transformed discharge summary moved from an

11th-grade to a 6th-grade reading level, with measured understandability rising from 13% to 81%. In the same study some errors were introduced, which is why the design below puts a clinician gate in the middle rather than shipping straight to the patient.

Approach

16. Ingest and structure. Take a PDF discharge summary or consent form and pull out the parts a patient actually acts on: medicines and changes, warning signs, follow-up appointments, activity limits, and who to call.

17. Rewrite with a clinician gate. Generate a plain-language version at a target reading level, then show it beside the original for a nurse or clinician to approve or edit before release. The gate is the safety design, and it is worth saying out loud in the pitch.

18. Visual breakdown. A medication timeline, a red-flag card of symptoms that mean call someone now, and a follow-up calendar. Most of the comprehension gain comes from structure, not from prose.

19. Questions on the document. A question box scoped strictly to the patient's own record, answering only from the approved text and saying so when the answer is not there. This is the smallest part of the build and should be the smallest part of the pitch.

20. Language and format options. Te reo Māori, Pacific languages, large print, and audio. This is where Right 5 is actually met, and it is the part no overseas product covers for New Zealand.

Positioning

Health chatbots are the most crowded category at SDG hackathons. If you lead with the chat box you will look like several other teams. Lead with the consent form and the legal right, treat the question box as a supporting feature, and the entry reads differently.

Build and demo

Why 48 hours works PDF text extraction, one structured extraction call, one rewrite call against a reading-level target, and a review screen. No training and no integration with any hospital system. The review interface is ordinary CRUD.

Demo Put a real consent form on screen beside the rewritten version, with the reading grade shown for each. Then have the clinician view catch and correct one generated line — demonstrating the gate working is more convincing than pretending the model is always right.

Risks

- Never use real patient records. Use synthetic or published sample documents only, and say so.
- The model will occasionally introduce errors. Do not claim otherwise — build the review step and make it part of the demo.
- Translation into te reo Māori and Pacific languages needs care. Show it as a designed pathway with human verification rather than claiming machine translation is sufficient.
- Position it as supporting the consent conversation, not replacing it.
- Strong legal hook and clear evidence behind it. The risk is presentation — pitched as a chatbot it blends in, pitched as a Right 5 tool it does not.

Idea 7: Disaster Recovery & Sudden Financial Shocks

The Issue: When extreme weather or economic shocks hit vulnerable communities, relief agencies take weeks to identify who needs immediate micro-grants or emergency support most urgently.

Software Solution: A predictive vulnerability mapping dashboard that combines satellite imagery (flood/drought data) with open economic indicators to help aid organizations auto-allocate emergency micro-disbursements to affected households instantly.

Comparison

#	Idea	SDG	Novelty	48h risk	Emotion	Tech
1	Cold-chain decisions	3.b.1	Medium *	Low	High	Med
2	Grower surplus matching	12.3 + 2.1	High	Low	High	Med
3	Disability funding tool	10.2	High	Low	Highest	Med
4	E-waste routing	12.5	Med-high	Low	Low	Med
5	Indicator generator	11.7.1	High	High	Medium	High
6	Health literacy and consent	3.8.1 + 10.2	Medium †	Lowest	High	Low

* Medium only if pitched as the decision layer. Low if pitched as a logger, since ColdTrace X already covers that.
† Medium because Epic already ships plain-language result summaries. The open ground is the full discharge summary, the consent form, and te reo Māori and Pacific language output, none of which overseas products cover for New Zealand.

Suggested picks

Strength	Pick
Best all round	Idea 2 — Grower surplus matching
Best local resonance	Idea 3 — Disability funding tool
Best technical showing	Idea 5 — Indicator generator
Safest hardware demo	Idea 1 — Cold-chain, reframed
Easiest to finish well	Idea 6 — Health literacy and consent

Idea 6 is the lowest-risk build on the list and the easiest to demo convincingly, but it sits in the most crowded category. It rewards a team that is stronger on presentation than on engineering, provided the pitch leads with the Code of Rights rather than the chat box.
Worth weighing: the probe-in-ice moment in Idea 1 is the most memorable few seconds available to any team in the room, and a web app cannot match it. If someone on the team does embedded work, keep the prop — just make sure the software behind it is the decision layer.

Not included

Rural volunteer transport matching was considered and dropped. Road XS, Spedsta, and Assisted Rides already do it, and Hato Hone St John and the Cancer Society already run the NZ services. Where it has been studied, the bottleneck is human coordination, not the matching software.

Practical notes

Teams split at noon on day two — half keep building, half prepare the pitch. Agree who is on which side beforehand. The 30 minutes between the end of the hack and the start of pitching is the only window to regroup.
Pitch structure, from the official template:

Time	Section	Content
0:30	Connect	Who you are, what you care about

1:00	Problem	Who is affected, made human, with data
2:00	Big idea	How it solves the problem, plus the prototype
1:30	Impact	What changes, what you need, what it means longer term

The template notes: describe the problem you are solving, which may not be the one you set out to solve.

Sources

SDG

UN SDG Report 2026 — <https://unstats.un.org/sdgs/report/2026/>

SDSN Report 2026 — <https://dashboards.sdgindex.org/>

UN-Habitat SDG 11 Report 2026 —

<https://unhabitat.org/sdg-11-global-report-2026-housing-at-the-centre-of-sustainable-cities-and-communities>

Cold chain

Health NZ cold chain standards —

<https://www.tewhātuora.govt.nz/health-services-and-programmes/vaccine-information/vaccine-service-delivery/cold-chain-standards-for-vaccines/>

National Standards 2017 (2nd ed) —

<https://www.tewhātuora.govt.nz/assets/For-the-health-sector/Health-sector-guidance/Vaccine-information-for-healthcare-professionals/National-Standards-for-Vaccine-Storage-and-Transportation-for-Immunisation-Providers-2017-2nd-edition.pdf>

IMAC cold chain — <https://immune.org.nz/vaccines/cold-chain>

Nexleaf ColdTrace — <https://www.nexleaf.org/work/vaccines/>

Food

NZFN Hunger Monitor 2025 —

<https://www.nzfoodnetwork.org.nz/wp-content/uploads/2026/03/NZFN-Hunger-Monitor-2025-Digital.pdf>

NZFN impact stats — <https://www.nzfoodnetwork.org.nz/our-impact/>

NZFN and Lineage partnership —

<https://www.nzfoodnetwork.org.nz/food-rescue-partnership-nz-food-network-lineage-grant/>

Rabobank food waste — <https://www.rabobank.co.nz/community/food-waste>

PMCSA food rescue evidence — <https://www.pmcsa.ac.nz/topics/food-rescue-food-waste/>

Disability

DSS February 2026 fact sheet —

<https://www.disabilitysupport.govt.nz/disabled-people/improvements-to-disability-support-services/february-2026-fact-sheet>

NZ Disability Strategy 2026–2030 —

<https://www.whaikaha.govt.nz/assets/NZDS-26-30-Documents/NZDS-2026-2030.pdf>

CCS Disability Action, Budget 2026 —

<https://www.ccsdisabilityaction.org.nz/news/budget-2026-statement>

Blind Low Vision NZ submission —

<https://blindlowvision.org.nz/news/submission-on-the-disability-support-services-bill-2026/>

E-waste

TechCollect NZ stewardship — <https://techcollect.nz/e-waste-product-stewardship-new-zealand/>

Electronic waste in New Zealand — https://en.wikipedia.org/wiki/Electronic_waste_in_New_Zealand

Data and tooling

data.govt.nz APIs — <https://data.govt.nz/catalogue-guide/using-data-govt-nz-apis>

NZ AI Strategy — <https://www.mbie.govt.nz/assets/new-zealands-strategy-for-artificial-intelligence.pdf>

Health literacy and consent

HDC Code of Rights (Rights 5 and 6) —

<https://www.hdc.org.nz/your-rights/about-the-code/code-of-health-and-disability-services-consumers-rights/>

Code of Rights, full PDF —

https://www.hdc.org.nz/media/550hs5ih/code-of-rights_online_5-sept-2022.pdf

HQSC health literacy background —

<https://www.hqsc.govt.nz/assets/Consumer-hub/Partners-in-Care/Publications-resources/health-literacy-background-info-Sep-2013.pdf>

Health literacy and Pacific peoples review — <https://ojs.aut.ac.nz/pacific-health/article/view/4>

NZ Society of Anaesthetists on health literacy —

<https://anaesthesia.nz/news/health-literacy-in-aotearoa-new-zealand-remains-a-critical-challenge/>

AI-generated patient-friendly discharge summaries —

<https://www.medrxiv.org/content/10.1101/2025.07.04.25330804.full.pdf>

NYU Langone discharge summary trial — <https://clinicaltrials.gov/study/NCT06711458>

Epic Emmie, patient-facing AI — <https://www.epic.com/software/emmie/>

Acute readmissions to hospitals, MoH —

<https://www.health.govt.nz/statistics-research/system-monitoring/planning-and-performance-data/acute-readmissions-to-hospitals>

Judging

GitLab AI Hackathon 2026 winners —

<https://about.gitlab.com/blog/gitlab-ai-hackathon-2026-meet-the-winners/>

Technical notes

data.govt.nz documentation references CKAN 2.7, which is old. Check /api/3/action/status_show before coding against those docs. The DataStore API only exists where a clean CSV was supplied; many listings are metadata-only links to agency sites.

Claude Sonnet 5 pricing rises to \$3/\$15 per million tokens after 31 August 2026.